A rising biocapacity total can mask ecological-capital land drawdown: a two-land model

Amin Abaee[a,*]

[a]*Independent Researcher, Tehran, Iran*

Abstract

Biocapacity combines land area and productivity in an aggregate annual account. We examine the information retained by this total using a conserved-area two-land model with delayed ecological regeneration, demographic response and an imposed quality constraint. Conversion increases the model account when the receiving land class has the higher per-hectare contribution, even as ecological-capital land declines. In a representative simulation, biocapacity increases 39%, capital-land area decreases approximately 46%, and its modeled account contribution decreases 51.72%. Sharp static bounds quantify how independent land-area and quality observations restrict the compatible allocation set. Integrated contribution budgets distinguish conversion, maintenance and retirement, restoration, quality and fast-yield effects. In the 100-year fixed-demand simulation, conversion contributes 0.547925 gha, while the net account increase is 0.231310 gha; other contributions largely offset conversion. Budget closure improves under timestep refinement. The reduced fixed-demand equality relation reaches the minimum-area boundary without a fold or critical slowing; this result is specific to that comparison. Separately, National Footprint Accounts Edition 2026 values for 1961–2023 show world biocapacity increasing 24.86%, biocapacity per person decreasing 52.71%, and production footprint per person increasing 16.73%. Cropland, forest and built-up accounts contribute +24.06, −5.95 and +7.44 percentage points, respectively, to total biocapacity change. These are annual account changes rather than physical-stock observations. The results support reporting constituent accounts alongside independently measured ecological assets.

Keywords: biocapacity, ecological footprint, land-use conversion, aggregation,

*Corresponding author. Email: amin_abaee@ut.ac.ir. ORCID: 0000-0002-0019-1842.

identifiability, critical slowing down, delay differential equations, carrying capacity

1. Introduction

In ecology, *carrying capacity* describes the population a resource base can sustain. Many models treat it as a fixed ceiling, in part because it is difficult to estimate (Cohen, 1995). If resource use changes the basis of future production, however, a fixed ceiling may not capture that change. An aggregate production-based measure may increase despite a decline in the area of a constituent land class.

The National Footprint and Biocapacity Accounts (NFA) report biological supply as biocapacity and human demand as the Ecological Footprint. Both are expressed in global hectares (gha), which adjust area for biological productivity relative to the world average in a given year; the annual basis is implicit (Global Footprint Network, 2009, 2019). Because a global hectare combines area and productivity, a change in the total does not by itself show whether yields, area, or the mix of land uses changed. Distinguishing these mechanisms requires information on the constituent areas and their per-hectare contributions.

In composite-indicator theory, this ambiguity is called *compensation*: a gain in one component can offset a loss in another (Nardo et al., 2008; Becker et al., 2017; Fischer et al., 2022). The aggregate remains useful, but reading it alone can conceal component changes. Related critiques discuss the interpretation of the Ecological Footprint and biocapacity accounts (Blomqvist et al., 2013; Giampietro and Saltelli, 2014; van den Bergh and Grazi, 2015; Galli et al., 2016).

Most human–environment models summarize nature as an aggregate stock or a unified feedback. Examples include the *World3* simulations (Meadows et al., 1972, 1974, 2004), models of Easter Island (Brander and Taylor, 1998; Anderies, 2000), and coupled Earth-system frameworks (Motesharrei et al., 2016; Donges et al., 2017; Calvin and Bond-Lamberty, 2018; Henderson and Loreau, 2018). Other work debates whether human carrying capacity is fixed, variable, or useful as a concept (Cohen, 1995; van den Bergh and Rietveld, 2004; Simon, 1996; Meyer, 1994; Meyer and Ausubel, 1999; Dasgupta, 2021); global demographic assessments often project population separately from environmental constraints (United Nations, 2022). Land-use studies, meanwhile, document conversion among forests, grasslands, wetlands, and

cropland (Haberl et al., 2007; Krausmann et al., 2013; Erb et al., 2017). These studies motivate an analysis of the information retained by aggregate biocapacity when land classes differ in yield and regeneration.

Annual biological provision and the ecological assets supporting that provision are distinct quantities.

Analogies. To make the accounting issue concrete, imagine an orchard with fast-provisioning ground, such as intensively managed beds, beside slower-renewing ground with trees and living soil. Annual harvest is a *flow*; trees, roots, and soil are assets that support future flows. In our model, a shortfall can trigger a transfer from ecological-capital land to fast-provisioning land. If the receiving class has a higher annual contribution per hectare in that scenario, the conversion term adds to B at the margin while capital-land area A_c falls; other flows can offset that increment. This is a conditional model mechanism, not evidence that NFA accounts measure physical stocks or that a particular real-world conversion occurred. Replanting may restore a later harvest, but it is not a one-for-one measure of the condition or recovery of the distinct assets represented by the model's capital-land class. The analogy separates current output from the state of its productive base; it does not assert that the classes map directly to NFA categories.

A second analogy separates observed service from structural condition: an elevator rated for ten people might continue operating while carrying fourteen, but the ride itself does not establish whether repeated overloading has changed the cable's safety margin. Continued service is not a measurement of structural condition, nor does the analogy predict inevitable failure. Related material-ledger work discusses a broader flow–stock masking problem; here we study its composition-specific form in a biocapacity model (Abaee, 2026c).

Research questions. This paper addresses three questions:

1. **The masking dynamic:** Under what conditions can conversion raise total model biocapacity while A_c declines, how long can this divergence last, and can the aggregate identify the land shift?

2. **System trajectory:** Under the model's stated assumptions, does drawdown involve a fold, hysteresis, or critical slowing, or does the reduced comparison follow another path?

3. **Limits and recovery:** What limits drawdown under fixed demand, and what modelled process allows capital-land area to recover?

Core contributions. We make three contributions:

- We construct a dynamic two-land model that conserves total area, represents capital-land quality separately from area, and includes delays in ecological regeneration and demographic response.
- We derive the condition under which a conversion term raises aggregate biocapacity while A_c falls, quantify sharp static identification intervals under independent observation packages, and compute signed integrated contribution budgets with refinement and closure checks. In a reduced fixed-demand comparison, the balance relation reaches the area floor without a fold or critical slowing; realized paths in the delayed model remain history-dependent. These are conditional model results, not general claims about observed ecosystems.
- We decompose NFA Edition 2026 account values for 1961–2023 and examine a separate cropland-area/yield proxy. The account series show a compositional change, not a census of physical ecological stocks.

Finally, we derive five conditional predictions and specify their observation requirements and rejection criteria (SI §S11). The imposed-cap comparison is an internal consistency test rather than an ecological generalization. The model parameters are representative rather than fitted, so simulations illustrate conditional mechanisms rather than historical forecasts. Aggregate biocapacity is therefore most informative when read alongside its component accounts.

2. Methods

2.1. A two-land model with conserved area and ecosystem quality

To examine how an aggregate indicator can mask a change in composition, we divide total bioproductive area A_{tot} among three mutually exclusive classes:

- **Fast-provisioning land** (A_f): cropland and other intensively managed systems, represented by their annual contribution.

- **Ecological-capital land** (A_c): forest, soil, wetland, and long-rotation systems grouped into one effective class, rather than treated as a single real land type.
- **Reserve land** (A_r): fallow, retired, or not-yet-allocated area.

The three areas sum to A_{tot}. The model also assigns capital land a separate, uncalibrated, model-defined quality index $q \in [0, q_{\max}]$. This effective state changes the modeled annual contribution per hectare; it is not a direct measurement of biomass density, soil integrity, or ecological health. Larger q need not imply a larger annual account contribution: q is a regeneration/readout state, and the growth-linked part of the observation function is non-monotone. The baseline bound $q_{\max} = 1$ is imposed for admissibility, not estimated as an ecological threshold. The raw regeneration-rate function is logistic:

$$G_c(q) = \rho_c \, q \left(1 - \frac{q}{q_{\max}}\right), \tag{1}$$

Capital land's annual contribution per hectare combines a direct-yield term with a term linked to quality growth:

$$\Upsilon_c(q) = b_c + b_{G,c} \, G_c(q), \tag{2}$$

where b_c is direct yield and $b_{G,c}$ converts the quality-growth term into annual account value. The fast- and capital-land contributions, Y_f and Y_c, sum to total model biocapacity:

$$B = Y_f + Y_c = b_f A_f + \Upsilon_c(q) \, A_c, \tag{3}$$

Here b_f is fast-land contribution per hectare. Model B is an annual account-equivalent amount in gha, with the annual basis implicit (Global Footprint Network, 2009, 2019).

Demand, shortfalls, and land conversion. Total demand is $E = eP$, where P is population and e is the constant footprint per person (gha per person; cap denotes one person in the equations and tables). The shortfall S is the demand not met by the human-available shares of the two land contributions:

$$S = [E - \sigma_f Y_f - \sigma_c Y_c]_+ , \tag{4}$$

The positive-part operator is $[x]_+ = \max(x, 0)$; baseline availability fractions are $\sigma_f = \sigma_c = 1$. When $S > 0$, the model transfers capital land to fast-provisioning land at a finite rate:

$$u_c = \frac{1}{\tau_{\text{conv}}} \min\left(\frac{S}{b_f},\ A_c - A_c^{\min}\right) \qquad [\text{ha}\,\text{yr}^{-1}], \tag{5}$$

The ratio S/b_f is the fast-land area required to cover the shortfall under this model rule; $A_c^{\min}$ is the lower area bound, and τ_{conv} sets the conversion timescale. This transfer is a specified response rule, not a reconstruction of observed land-clearing decisions.

Land flows and asymmetric restoration. Land moves among classes through one-to-one area transfers:

$$\dot{A}_c = -u_c + R_{rc}, \tag{6}$$

$$\dot{A}_f = u_c + \mu A_r - \eta_f A_f, \tag{7}$$

$$\dot{A}_r = -\mu A_r + \eta_f A_f - R_{rc}, \tag{8}$$

The flow μA_r moves reserve area into fast use, while $\eta_f A_f$ retires fast land to reserve. Reserve land returns to the capital-land class only through the surplus-gated restoration flow R_{rc}:

$$R_{rc} = \chi_r\, A_r\, [B - E]_+ . \tag{9}$$

Restoration is therefore asymmetric in the model: fast land may be retired into reserve, but reserve returns to ecological-capital use only when the modeled surplus is positive and the restoration flow is active. A closed shortfall alone does not restore capital-land area.

Delays, model debt, and demographics. Quality responds to its delayed value through projection of the regeneration vector field onto the tangent cone of $K_q = [0, q_{\max}]$. Population responds to carrying capacity evaluated at $t - \tau_p$; the delay is in the capacity input, not in population itself in its crowding term. The model also tracks a leaky deficit state D that affects fast-land

yield:

$$\dot{q} = \Pi_{T_{K_q}(q(t))} G_c\big(q(t-\tau_g)\big), \tag{10}$$

$$\dot{P} = rP\left(1 - \frac{P}{K(t-\tau_p)}\right), \qquad K = \max(B/e,\, K_{\min}), \tag{11}$$

$$b_f = \big(b_{f0} + T_b(t)\big)\, e^{-\alpha D}, \tag{12}$$

$$\dot{D} = [E - B]_+ - \eta D, \tag{13}$$

$$T_b(t) = \omega_{Tb}\, \frac{\Delta b}{1 + e^{-\kappa_w (t - t_{wave})}}, \qquad \omega_{Tb} \in \{0, 1\}. \tag{14}$$

Here $K_q = [0, q_{\max}]$, $T_{K_q}(q)$ is its tangent cone at the current state, and Π is Euclidean projection onto that cone. We interpret the quality equation as an absolutely continuous solution, almost everywhere, with $\rho_c > 0$ and an admissible continuous history $q(s) = \phi(s) \in K_q$ for $s \in [-\tau_g, 0]$. Since $G_c(x) \geq 0$ for $x \in K_q$, the projected law agrees with the raw delayed rate below the upper bound and sets outward growth to zero at $q = q_{\max}$; at $q = 0$ the vector field is nonnegative. The cap is a mathematical admissibility constraint, not an empirically identified biological saturation mechanism. Areas A_f, A_c, and A_r are measured in hectares; B, E, and S are annual account amounts in gha, with the annual basis implicit. D is measured in gha yr; both $\dot{D}$ and ηD are in gha. It is an internal model state driven by aggregate deficit, not an area, a physical stock, or an NFA-reported quantity. The optional technology boost T_b raises fast-land yield along a bounded logistic curve. It is off in the baseline, figures, and tables, and appears only in the matched comparison in SI §S8. The dynamic states are A_f, A_c, q, P, and D; timber liquidation and direct restoration from fast land to capital land are not active in the reported runs (SI §S1).

Core system feedbacks. The equations permit two conditional feedbacks:

1. **Compositional-yield feedback.** When fast land contributes more per hectare than capital land ($b_f > \Upsilon_c(q)$), conversion raises B at the margin while reducing A_c. With endogenous population, higher B can raise delayed carrying capacity K and later population and demand, potentially creating another shortfall. Maintenance, retirement, quality change, and the other model terms also affect B; this is a model feedback, not an empirical causal finding.

2. **Deficit–yield feedback.** A shortfall adds to D, while repayment draws it down. As D grows, modeled fast-land yield b_f falls; this can reduce B and deepen the shortfall. The state D is the model's leaky deficit measure, not a direct measure of nutrient depletion, aquifer overdraft, or other physical processes.

2.2. Parameter selection and empirical grounding

Table 1 summarizes the baseline parameter values and their evidence status. Parameters are representative rather than econometrically fitted; the model is designed to isolate structural mechanisms, not to generate precise historical forecasts. To keep the model tractable, it consolidates forests, soils, wetlands, and other slow-renewing systems into one effective capital-land class, although their extent, condition, yields, and recovery dynamics differ substantially. Published forest, fish-stock, and soil studies use heterogeneous recovery definitions and document system-specific responses (SI §S5.5) (Poorter et al., 2016; Hutchings and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011). We do not refit a common recovery curve or compute harmonized t_{50} or τ_r values from those records here. They are measurement and future-modeling precedents only; they do not estimate ρ_c or τ_g, validate the present generic quality equation, or identify its parameters.

Two data-based quantities provide context for cropland but do not calibrate the model: the NFA cropland account value per unit area and a separate cereal-yield/cropland-area proxy (SI §S5). Their boundaries do not map one-to-one onto model fast land, and neither is a fitted value of b_f or a direct measure of ecological capital. Finally, the baseline assumes $b_f - \Upsilon_c(q) > 0$. This is an explicit conditional modeling regime for examining when a short-term contribution premium can mask a change in land composition, not a universal empirical claim.

2.3. Data and empirical measures

Our empirical analysis uses the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). We analyse complete world observations for 1961–2023 and exclude preliminary estimates for 2024–2025. A 1961–2022 sensitivity check uses the same account edition and population series, so the data vintage is held constant across the two windows.

From the workbook's `world_data` sheet, we extract total biocapacity B (`biocap_total_gha`), the Global Ecological Footprint of Production E_{prod}

Table 1: Baseline parameter values and evidence status. Representative values are effective, lumped quantities; the NFA account/area quotient is descriptive, not a physical-yield measurement. Extension parameters are active only in scenarios specified in this table or the SI. Symbols and units are listed in SI §S1.

Parameter	Value	Meaning	Status
ρ_c	$0.08\,\mathrm{yr}^{-1}$	quality-regeneration scenario rate	scenario setting; not estimated
b_{f0}	$0.85\,\mathrm{gha\,ha}^{-1}$	base fast-land yield per hectare	representative (2026 NFA account value/area quotient ≈ 1.045 in 1961; not a physical-yield measure)
b_c	$0.05\,\mathrm{gha\,ha}^{-1}$	capital-land direct yield	representative
$b_{G,c}$	$0.8\,\mathrm{gha\,yr\,ha}^{-1}$	regeneration-value coefficient for capital land	representative
$q_{\max}$	1.0 (dimensionless)	imposed quality admissibility bound	not an estimated ecological threshold
μ	$0.06\,\mathrm{yr}^{-1}$	cropland maintenance (reserve→fast)	representative
η_f	$0.05\,\mathrm{yr}^{-1}$	fast-land retirement (fast→reserve)	representative
τ_{conv}	1.0 yr	conversion time constant	representative
α	$0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	degradation of b_f per unit debt	representative
η	$0.05\,\mathrm{yr}^{-1}$	debt-repayment rate (> 0)	representative
e	$0.55\,\mathrm{gha\,cap}^{-1}$	per-capita footprint (constant)	representative
r	$0.02\,\mathrm{yr}^{-1}$	intrinsic population growth	representative
$\tau_g,\ \tau_p$	20, 25 yr	regeneration and demographic delay scenarios	scenario settings; τ_p lumps institutional lags
A_{tot}	2.5 ha	total bioproductive area	normalization
$\sigma_f,\ \sigma_c$	1, 1 (dimensionless)	human-available shares of annual account amounts	baseline (< 1: reservation extension)
χ_r	$0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$	surplus-gated restoration coefficient	on with endogenous demand; off under fixed demand
$A_c^{\min}$	0.05 ha	capital-land area floor	enforced by conversion-rate law
$A_r^{\min}$	0.10 ha	reserve monitoring threshold	not enforced by the update
$K_{\min}$	0.02 cap	carrying-capacity floor	imposed in definition of K
$\Delta b,\ \kappa_w,$ t_{wave}	$1.0\,\mathrm{gha\,ha}^{-1}$, $0.1\,\mathrm{yr}^{-1}$, 40 yr	technology wave	off in baseline; active only in the matched SI S8 experiment

(`efp_total_gha`), and UN PopStat population P (`pop_1000_persons`). Population is converted from thousands to persons. Per-person measures are calculated from unrounded totals rather than the workbook's rounded per-person columns; ratios, indices, and log changes are dimensionless.

Accounting categories and physical stocks. We retain the five workbook labels: *Cropland*, *Grazing land*, *Forest biocapacity*, *Fishing grounds*, and *Built-up land*. These are account values, not measurements of biomass, soil condition, wetland extent, fish biomass, or model area A_c. Built-up land is an NFA accounting class, not ecological capital. We therefore interpret movements among categories as changes in account composition, not as direct observations of physical ecosystems or of the model's capital-land stock.

Auxiliary cropland proxy. As a separate comparison, we pair Our World in Data's FAOSTAT-derived world cropland-area series (arable land plus permanent crops) with cereal yield (Our World in Data, 2026b,a). Cereal yield is a limited productivity indicator, not a measure of all-crop output or an NFA yield factor. We scale the proxy to the NFA cropland share of total biocapacity in 1961, calculate the crop-account proxy $X_t = \hat{b}_{f,t} A_{f,t}$, and define the remainder as $C_t^{\mathrm{emp}} = B_t - X_t$. Here $A_{f,t}$ denotes observed cropland area for this auxiliary calculation, not the model state. The scaling makes this an assumed account split, not an independent measurement of the model's fast-land contribution or of physical capital. The remainder absorbs differences in boundaries, categories, and calibration; it is not $\Upsilon_c(q)A_c$. SI §S5 reports the proxy and its calibration sensitivity. We denote this empirical remainder by C^{emp}, distinct from the model observation residual $C^{\mathrm{mod}} = B - b_f A_f = \Upsilon_c(q)A_c$. We infer no physical-stock trend from it.

Total and per-person change. We use the exact identity

$$\Delta \ln B = \Delta \ln(B/P) + \Delta \ln P$$

to separate the scale change in total biocapacity from the change in biocapacity per person. This is an accounting identity, not a causal model of biocapacity production.

2.4. Numerical implementation

Fixed-demand intervention.. In the controlled fixed-demand runs, we override $E = eP$ with $E(t) = \bar{E}$ and set $R_{rc} = 0$. Population still follows its

displayed demographic equation, but no longer determines demand in that intervention. Thus fixed demand and restoration-off are two explicit scenario settings, not consequences of demographic equilibrium. Endogenous-demand restoration runs retain $E = eP$ and the stated restoration gate.

We integrate the delay equations by the method of steps with an explicit first-order Euler scheme. At step k, current states are read at t_{k-1}; time-dependent inputs and delayed values are evaluated at t_k, with linear interpolation of the delayed q and B histories. The raw delayed quality velocity is projected onto $T_{K_q}(q(t_{k-1}))$ before the Euler update; the updated quality is then clipped to K_q as a finite-step safeguard. The explicitly enforced state bounds are applied, reserve area is reconstructed as $A_r = A_{tot} - A_f - A_c$, and the updated B is stored in the delay history. The reserve threshold $A_r^{\min}$ is monitored rather than projected; the optional $A_f^{\max}$ ceiling is not active in reported runs. The positive-part operator is evaluated exactly. Baseline demonstration and high-resolution threshold runs use $\Delta t = 0.05\,\mathrm{yr}$; broader drawdown sweeps use $0.1\,\mathrm{yr}$, with selected thresholds rechecked at $0.05\,\mathrm{yr}$.

Pre-run histories are initialized over the full longest active delay window, $[-\max\{\tau_g, \tau_p\}, 0]$, as $q(s) = q_0$ and $B(s) = B_0$, with B_0 calculated from the initial state. In the matched technology-wave experiment (SI §S8), both runs share the no-wave prehistory; the active $T_b(0)$ changes current $B(0)$ and subsequent wave-adjusted $B(t)$ enters the delayed carrying-capacity input $K(t - \tau_p)$. The enforced state bounds are $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in K_q$, $P, D \geq 0$, and $K \geq K_{\min}$. The reserve threshold $A_r^{\min} = 0.10\,\mathrm{ha}$ is a post-hoc monitoring criterion, not an invariant enforced by the update; $A_f^{\max}$ is not imposed in the reported runs.

A finite-step consistency test compared post-Euler quality clipping with explicit tangent-cone projection before the Euler update. At $\Delta t = 0.05\,\mathrm{yr}$ and $T = 100\,\mathrm{yr}$, using the Figure 1 initial state and parameters, the tested arrays for A_f, A_c, q, P, D, A_r, B, E, and T_b were exactly equal at $\tau_g = 5, 20, 40\,\mathrm{yr}$. For the continuous baseline quality path at $\tau_g = 20$ or $40\,\mathrm{yr}$, $q(t) = \min(1, 0.9 + 0.0072t)$; its maximum piecewise-linear error fell from 7.1×10^{-5} at $\Delta t = 0.1\,\mathrm{yr}$ to 8.9×10^{-6} at $0.0125\,\mathrm{yr}$. In the full fixed-demand Figure 1 case ($P_0 = 1.5\,\mathrm{cap}$), halving Δt from 0.05 to $0.025\,\mathrm{yr}$ reduced maximum absolute differences against a $0.003125\,\mathrm{yr}$ reference from 0.00221 to $0.00102\,\mathrm{ha}$ for A_c, 0.00180 to $0.000831\,\mathrm{gha}$ for B, 0.000245 to $0.000114\,\mathrm{cap}$ for P, and 0.00200 to $0.000924\,\mathrm{gha\,yr}$ for D (SI §S3). These checks establish equality only for the enumerated cases and show smaller differences under timestep refinement; they are not a universal error bound

or proof of convergence for all settings. The contribution-budget calculation uses seven nested steps through 0.00078125 yr and independent RHS-based quadrature (Section 3.6; SI §S10). The numerical implementation evaluates the contribution integrands separately from the endpoint account change.

All reported baseline trajectories use the projected delayed-quality law in Eq. (10). SI §S3 presents the current-quality comparator $G_c(q(t))$ only as a sensitivity check; it is not the selected model.

3. Results

3.1. When conversion raises the total as capital land declines

State versus annual readout.. The term $b_{G,c}G_c(q)$ values a logistic growth flow, not a stock level. Consequently, $\Upsilon_c(q) = \Upsilon_c(q_{\max} - q)$: two distinct interior states can have the same per-hectare account value. Above $q_{\max}/2$, the state can increase while this growth-linked readout falls; at the cap only the direct-yield term b_c remains. For the baseline path, q rises from 0.9 to 1, whereas Υ_c falls from 0.05576 to 0.05 gha ha^{-1}. This is an intentional growth-flow observation rule, not evidence that improved ecological condition reduces physical production. A physical interpretation of q would require independent condition measurements and validation of this observation rule.

For a given capital-land quality level q, its annual contribution per hectare, $\Upsilon_c(q)$, is a concave quadratic. It peaks at $q^* = q_{\max}/2$; with the baseline parameters, the maximum is 0.066 gha ha^{-1}. The quality-related term contributes 0.016 of that value, while direct yield contributes 0.05 gha ha^{-1}; the direct-yield component is larger in this case.

Conversion changes the aggregate according to the gap between the annual account contribution of one fast-land hectare and the effective value of one capital-land hectare. We call this gap the *composition diagnostic*:

$$\Delta b_{\text{conv}}(q, D) = b_f(D) - \Upsilon_c(q). \tag{15}$$

At baseline, Δb_{conv} equals +0.787, +0.785, and +0.794 gha ha^{-1} at $q = 0.3$, 0.6, and 0.9 (rounded to three decimals). Each lies in the composition-premium regime ($\Delta b_{\text{conv}} > 0$), where conversion adds to the aggregate at the margin. At $\Delta b_{\text{conv}} = 0$ the contribution is first-order neutral; below zero, it is a composition-discount. These labels describe the conversion term only. Maintenance, retirement, quality change, and other flows still determine the net change in B. At any given quality level, that contribution changes linearly with A_c; it has no interior maximum with respect to area.

Proposition 1 (Conversion, aggregate change, and what the total can reveal)**.** *Let $T_b \equiv 0$, $R_{rc} \equiv 0$ and hold D fixed, so that b_f is constant. Along any solution of (3)–(10),*

$$\dot{B} = \Delta b_{\text{conv}}(q)\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}\, G_c'(q)\, A_c\, \dot{q}. \tag{16}$$

(i) During conversion ($u_c > 0$), A_c falls when restoration is off. The conversion component of $\dot{B}$ is $\Delta b_{\text{conv}}(q)u_c$, positive exactly when $\Delta b_{\text{conv}}(q) > 0$. When maintenance balances retirement ($\mu A_r = \eta_f A_f$) and quality is unchanged ($\dot{q} = 0$), it is the only term in (16) and determines the sign of net $\dot{B}$. These restrictions are sufficient, not necessary; otherwise the net sign depends on the magnitudes of all active terms, including the conversion contribution.

(ii) At any instant in the reduced comparison—with fixed demand and debt, baseline availability shares, balanced maintenance and retirement, and no quality change—a positive Δb_{conv} makes the shortfall shrink while conversion is active. This is a pointwise statement: with $R_{rc} = 0$, $M = \mu A_r - \eta_f A_f$ evolves as $\dot{M} = -(\mu + \eta_f)M - \eta_f u_c$, so $M = 0$ is not invariant when $u_c > 0$. The pointwise identity therefore does not imply that the shortfall closes or that the trajectory reaches the area floor under the displayed land flows.

(iii) At fixed $b_f > 0$ and q, $B = b_f A_f + \Upsilon_c(q)A_c$ supplies one equation for the two land areas. Near an interior state, a small change in A_c can be offset by $\delta A_f = -(\Upsilon_c/b_f)\delta A_c$ and $\delta A_r = -\delta A_f - \delta A_c$, preserving both B and total area as long as the land bounds are respected. Thus several nearby allocations can have the same B; the aggregate alone does not identify A_c. For a finite observed change ΔB, the offset is $\Delta A_f = (\Delta B - \Upsilon_c \Delta A_c)/b_f$. It is feasible only if all resulting areas remain within bounds, so the feasible range depends on the starting state. If A_f and b_f are observed independently, the residual $C^{\text{mod}} = B - b_f A_f = \Upsilon_c(q)A_c$ identifies their product, not its two factors. An independent observation of A_c would give $\Upsilon_c(q) = C^{\text{mod}}/A_c$. From the supply observation relation alone, a single Υ_c value cannot separate b_c, $b_{G,c}$, ρ_c, and q. Observations of $\Upsilon_c(q)$ at multiple interior q values constrain b_c and the product $b_{G,c}\rho_c$ once $q_{\max}$ is specified. This does not rule out dynamic identification of ρ_c from a calibrated quality trajectory with known delay and history and interior observations; that would use (10), not the supply relation alone.

Proof. Differentiate (3) with b_f constant and substitute (6)–(7) with $R_{rc} = 0$: $\dot{B} = b_f(u_c+\mu A_r-\eta_f A_f)-\Upsilon_c u_c+b_{G,c}G_c'(q)A_c\dot{q}$, which is (16). (i) Follows from (16) with $R_{rc} = 0$; when the maintenance–retirement imbalance and quality-growth terms vanish, only the conversion term remains. (ii) At an instant satisfying its restrictions, $S = E - B$ and $\dot{S} = -\dot{B} = -\Delta b_{\text{conv}} u_c < 0$. With $M = \mu A_r - \eta_f A_f$ and $R_{rc} = 0$, Eqs. (6)–(8) give $\dot{M} = -(\mu + \eta_f)M - \eta_f u_c$; hence $M = 0$ is not maintained under active conversion. This pointwise identity makes no claim that conversion closes the shortfall before the area floor binds. (iii) At fixed b_f and q, the observation map is linear; an interior feasible point admits the stated small perturbations along a level set of B, while finite changes must also obey the area bounds. The residual C^{mod} identifies only $\Upsilon_c(q)A_c$ unless A_c is independently observed; the parameter combination follows from the definition of $\Upsilon_c(q)$. □

Figure 1a follows the baseline run with demand fixed at $E = 0.825\,\text{gha}$ and restoration off. Initial states are $A_f = 0.6\,\text{ha}$, $A_c = 1.5\,\text{ha}$, $A_r = 0.4\,\text{ha}$, and $q = 0.9$. During conversion, model biocapacity B rises from 0.594 to 0.825 gha (39%); capital-land area falls from 1.500 to 0.808 ha (46%), while fast-land area grows from 0.600 to 0.923 ha.

Quality q rises from 0.9 to 1.0, moving farther above $q^* = 0.5$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to $0.05000\,\text{gha}\,\text{ha}^{-1}$. The modeled capital-land contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (51.7%)—more than the 46.2% area decline because value per hectare also decreases. The one-to-one transfers conserve area: $\Delta A_f + \Delta A_c + \Delta A_r = 0$.

The adjustment is fastest early and then slows. On the deficit side, the approach to balance is exponential, with slow rate $|\lambda_1| = 0.054\,\text{yr}^{-1}$ and an e-folding time of about 19 yr. By year 25, 98% of the rise in B and 85% of the decline in A_c have occurred ($B = 0.820\,\text{gha}$, $A_c = 0.913\,\text{ha}$); by year 50, 97% of the area decline is complete. At the reported numerical resolution, $\dot{B} > 0$ and $\dot{A}_c < 0$ together for about 149 yr.

Reserve adjustment creates the long tail. As fast land retires to reserve, its contribution falls and a small shortfall reappears. Conversion then continues until $A_r/A_f = \eta_f/\mu$. The duration is not a fixed horizon: the slow rate λ_1 sets its timescale, while Δb_{conv} and the initial shortfall affect its size (SI §S2.2).

The aggregate has a clear identification limit: B alone cannot show whether A_c is shrinking while another component grows. Independent observations of fast-land area A_f and yield b_f would let us calculate $C^{\text{mod}} =$

$B - b_f A_f$. But C^{mod} identifies only the capital-land contribution, not its area and per-hectare value separately.

We call a trajectory with rising B and falling A_c a *composition mask*. A related but distinct pattern is the *liquidation mask*, in which demand E stays fixed while A_c falls. The first compares aggregate biocapacity with land area; the second compares demand with land area. They can occur together, but neither implies the other. The diagnostic Δb_{conv} gives the sign of conversion's marginal contribution, not necessarily the net change in B when other terms are active; a liquidation mask does not require $\Delta b_{\text{conv}} > 0$. Observing B, E, and A_c distinguishes their defining patterns. Here, "liquidation mask" names a trajectory; it is not the optional timber-liquidation extension, which is off in every reported run.

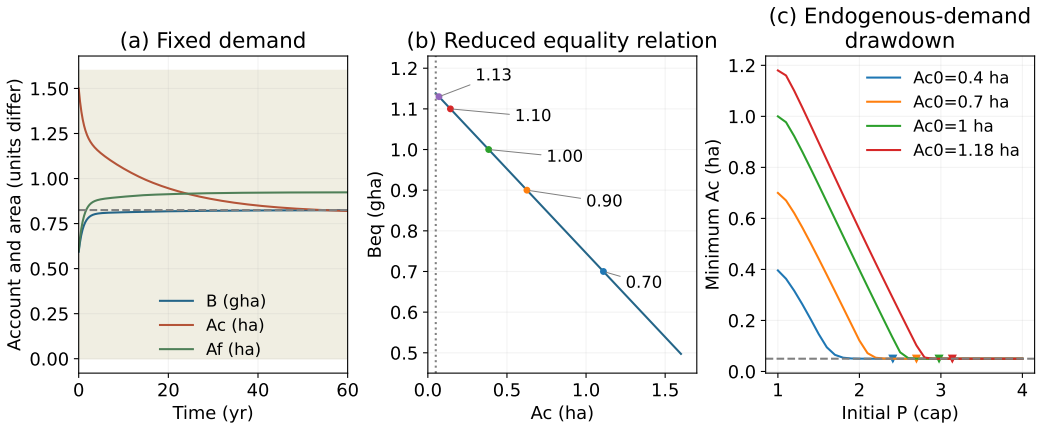


Figure 1: Baseline model results (Table 1). Account amounts B and E are in gha, land areas are in ha, and population is in cap. (a) Fixed demand ($E = 0.825$ gha), restoration off, and initial conditions $A_{f0} = 0.6$ ha, $A_{c0} = 1.5$ ha, and $q_0 = 0.9$. B rises from 0.594 to 0.825 gha as A_c falls from 1.500 to 0.808 ha and A_f rises from 0.600 to 0.923 ha. Shading marks the interval with $\dot{B} > 0$ and $\dot{A}_c < 0$ (about 149 yr; only the first 60 yr are shown). Within 25 yr, 98% of the rise in B and 85% of the fall in A_c occur. (b) Fixed-demand balance relation $B_{eq}(A_c)$; the listed balance areas are $A_c^*(E) = 1.110$, 0.626, 0.385, 0.143, and 0.070 ha at $E = 0.70$, 0.90, 1.00, 1.10, and 1.13 gha. The curve meets the minimum $A_c = 0.05$ ha at $E_{\text{ceil}} \approx 1.138$ gha; no balance point is admissible at higher demand. (c) Endogenous-demand runs with $\tau_g = 20$ yr, $\tau_p = 25$ yr, and $\chi_r = 0.10\,\text{gha}^{-1}\,\text{yr}^{-1}$. In the tested runs, the sampled minimum area varies monotonically with P_0 for each initial area A_{c0} ($E_0 = eP_0$); the finite grid does not establish a general continuity theorem. The floor is approached asymptotically. Triangles mark the proximity values in Table 2, defined as within 10^{-6} ha of the floor; no discontinuity is observed.

3.2. Fixed demand leads to a minimum-area boundary, not a fold

Once population reaches carrying capacity, $P = K = B/e$ and demand $E = eP$ equals biocapacity B; the shortfall then closes. Conversion is therefore transient in the endogenous-population runs. To isolate how the balance point changes with demand, we also fix E and analyze a reduced land system. The constraint $A_c \geq A_c^{\min}$ is a *typed floor*: it protects one land class, not the aggregate.

Proposition 2 (Balance point, demand ceiling and affine deficit-side dynamics)**.** *For this reduced comparison, hold demand at E, switch off the technology boost and restoration, set debt $D = 0$, and fix quality at $q = q_{\max}$. Under the projected law, $q_{\max}$ is the limit for an admissible history with $q(0) > 0$; the all-zero history remains at zero (SI §S2.3). Let $k = \mu/(\mu + \eta_f)$ and*

$$B_{eq}(A_c) = b_{f0}\, k\, (A_{tot} - A_c) + b_c A_c = b_{f0} k A_{tot} - (b_{f0} k - b_c) A_c. \tag{17}$$

(i) At a land-flow equilibrium with no conversion ($S = 0$) and $R_{rc} = 0$, reserve-to-fast maintenance balances fast-land retirement. Since $S = 0$ means $B \geq E$, it includes both the equality balance point and any feasible surplus allocation. The maintenance balance gives $A_f = k(A_{tot} - A_c)$ and $A_r = (1 - k)(A_{tot} - A_c)$, where $k = \mu/(\mu + \eta_f)$, and the resulting total is $B_{eq}(A_c)$. If $b_{f0} k > b_c$, this total decreases as A_c increases, so the equality equation $B_{eq}(A_c^*) = E$ has at most one solution:

$$A_c^*(E) = \frac{b_{f0} k A_{tot} - E}{b_{f0} k - b_c}. \tag{18}$$

The algebraic solution is feasible when $B_{eq}(A_{tot}) \leq E \leq E_{\text{ceil}}$, where $E_{\text{ceil}} = B_{eq}(A_c^{\min})$. It lies above the minimum area exactly when $E < E_{\text{ceil}}$. At lower demand, the algebraic root lies above A_{tot}, and every feasible no-conversion allocation already supplies at least E.

(ii) While shortfall is positive and conversion is not yet limited by the minimum-area constraint, the land subsystem is affine in $z = (A_f, A_c)$: $\dot{z} = Jz + c$, with

$$J = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix}, \qquad \det J = \frac{\mu + \eta_f}{\tau_{\text{conv}}} \Big(k - \frac{b_c}{b_f} \Big), \tag{19}$$

and $\operatorname{tr} J < 0$. *The matrix does not depend on demand or on the current state. Its determinant is positive exactly when* B_{eq} *decreases, and its fixed point is the balance point in (i). For* $B_{eq}(A_{tot}) \leq E < E_{\text{ceil}}$, *trajectories that remain in this regime approach that point at rates set by the eigenvalues of* J. *For* $E > E_{\text{ceil}}$, *the balance point would lie below the allowed minimum:* A_c *approaches* $A_c^{\min}$ *asymptotically and the shortfall tends to* $S^* = E - E_{\text{ceil}} > 0$. *Thus the reduced balance curve meets the minimum-area boundary; it does not turn back or form a fold. The leading eigenvalue does not approach zero at* $E = E_{\text{ceil}}$, *and the reduced equilibrium relation has no hysteresis. These conclusions hold under the stated assumptions and do not imply that the realized land allocation is independent of its history.*

(iii) Above the demand ceiling, with degradation active, debt approaches a unique value $D^*(E)$ *satisfying* $\eta D = E - B_{eq}(A_c^{\min}; D)$, *where* $B_{eq}(\cdot; D)$ *uses yield* $b_{f0}e^{-\alpha D}$. *This result requires* $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min})$; *the equilibrium debt increases with demand.*

Proof. (i) With $S = 0$ and $R_{rc} = 0$, (7) gives $\mu A_r = \eta_f A_f$, hence $A_r/A_f = \eta_f/\mu$ and, with $A_r = A_{tot} - A_f - A_c$, $A_f = k(A_{tot} - A_c)$; substituting into (3) with $q = q_{\max}$ ($\Upsilon_c = b_c$) and $D = 0$ gives $B = B_{eq}(A_c)$, and $S = 0$ requires $B \geq E$. Strict monotonicity and (18) follow from $b_{f0}k > b_c$. (ii) On the deficit side $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$ is affine in z; substituting into (6)–(7) with $A_r = A_{tot} - A_f - A_c$ gives J and a constant vector c by direct differentiation; $\det J$ follows by expansion, and $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f < 0$ for $b_c < b_f$. The fixed point solves $u_c = 0$ and $\mu A_r = \eta_f A_f$, i.e. it is the balance point. Because J is constant, convergence rates are constant along the branch. (iii) The map $D \mapsto \eta D + B_{eq}(A_c^{\min}; D) - E$ is continuous, negative at $D = 0$ when $E > E_{\text{ceil}}$, tends to $+\infty$, and is strictly increasing when $\eta > \alpha b_{f0} k (A_{tot} - A_c^{\min}) \geq -\partial_D B_{eq}$; hence it has a unique root, which increases with E. □

The equality point $B_{eq}(A_c^*) = E$ is unique when feasible, but at fixed exogenous demand every feasible maintenance-balanced allocation with $B_{eq}(A_c) \geq E$ has $S = 0$ and no conversion; strict-surplus allocations are a separate no-conversion family, subject to the land bounds. The deficit-side matrix J describes only active-shortfall dynamics, and the switch at $S = 0$ means its eigenvalues do not establish two-sided stability of an isolated equality point. In the full endogenous-population model, when $K_{\min}$ is inactive, a

positive stationary population also requires $E = B$, so its no-conversion equilibrium family lies on the equality relation with P adjusted to B/e. In the baseline run, $R_{rc} = 0$ and conversion has no reverse flow: after the shortfall closes, A_c stops falling and remains below its pre-overshoot level. Meanwhile, q evolves separately, so $\Upsilon_c(q)$ can still change. This is not perpetual decline at equilibrium and is not a claim about real ecosystems.

At baseline, $k = 0.5455$, the balance-branch slope parameter is $b_{f0}k - b_c = 0.414\,\mathrm{gha\,ha^{-1}}$, and the demand ceiling is $E_{\mathrm{ceil}} = 1.138\,\mathrm{gha}$. Equation (18) predicts balance areas of 1.110, 0.626, 0.385, 0.143, and 0.070 ha at demands of 0.70, 0.90, 1.00, 1.10, and 1.13 gha; these match the integrated equilibria (Fig. 1b). The eigenvalues of J are real and negative ($\lambda_1 = -0.0537$, $\lambda_2 = -0.998\,\mathrm{yr^{-1}}$). Across the tested fast-land yields, λ_1 shifts only from -0.0522 at $b_f = 0.60$ to $-0.0556\,\mathrm{yr^{-1}}$ at $b_f = 2.00$ and never changes sign (Table 2a).

Above the ceiling, using the same unrounded inputs in both calculations, $E_{\mathrm{ceil}} = 1.138409091\,\mathrm{gha}$, and $E = E_{\mathrm{ceil}} + 0.02 = 1.158409091$ and $E_{\mathrm{ceil}} + 0.10 = 1.238409091\,\mathrm{gha}$ give $D^* = 1.209655$ and $5.394538\,\mathrm{gha\,yr}$. Integration from the maintenance-balanced allocation at $A_c = A_c^{\min}$, with $D(0) = 0$, $q = q_{\max}$, $\Delta t = 0.1\,\mathrm{yr}$ and $T = 1200\,\mathrm{yr}$ gives the same values to the displayed precision (SI §S2.4). The uniqueness condition holds with margin: $\alpha b_{f0} k (A_{tot} - A_c^{\min}) = 0.034\,\mathrm{yr^{-1}}$, below $\eta = 0.05\,\mathrm{yr^{-1}}$. Under fixed demand, debt therefore saturates rather than growing without bound.

Critical slowing down refers to a fold approach in which the slowest return rate tends to zero and the return time $T_{ret} \approx -1/\lambda_1$ grows. That does not occur along this reduced balance branch: its linearization is constant, λ_1 stays away from zero, and the return time remains near 19 yr as E approaches E_{ceil}. The modeled signal is instead A_c approaching its minimum—information an aggregate-only observation cannot provide.

In the tested runs, the sampled minimum A_c varies monotonically with initial population P_0 for each starting area A_{c0} (Fig. 1c); the finite grid and selected bisections do not establish a general continuity result. The trajectory approaches, but does not cross, the floor because $u_c \leq (A_c - A_c^{\min})/\tau_{\mathrm{conv}}$. A larger initial capital-land area supports a larger P_0 before the run approaches that floor.

Table 2b gives the initial populations for which A_c comes within 10^{-2}, 10^{-4}, or 10^{-6} ha of its minimum. These tolerance-based values are proximity markers, not abrupt thresholds. At the tightest tolerance, $P_0 \approx 2.41$, 2.70, 2.98, and 3.14 cap for $A_{c0} = 0.40$, 0.70, 1.00, and 1.18 ha, respectively. They rise with starting area; no jump appears at any tested tolerance.

At $A_{c0} = 1.0\,\text{ha}$, the delayed-quality $10^{-6}\,\text{ha}$ proximity values shift by at most $0.001\,\text{cap}$ across the tested $\tau_g = 5, 20, 40\,\text{yr}$ cases (SI Table S2). Under the baseline constant history $q(s) = 0.9$ with $\rho_c = 0.08\,\text{yr}^{-1}$, the projected quality paths for $\tau_g = 20$ and $40\,\text{yr}$ are exactly identical: both reach the imposed ceiling at $125/9 \approx 13.8889\,\text{yr}$, before either delay elapses. Their matching outcomes are therefore a saturation-induced degeneracy, not evidence of general delay robustness; the 5 yr case is distinct. The restoration comparison is unchanged to three decimals because surplus-gated restoration is inactive while $B < E$ (SI §S4).

Table 2: Model results at the baseline parameter set. (a) At $q^* = q_{\max}/2$ ($\Upsilon_c(q^*) = 0.066\,\text{gha}\,\text{ha}^{-1}$), the table reports $\Delta b_{\text{conv}}(q^*)$, the eigenvalues of the deficit-side matrix J in (19), and the demand ceiling across a sweep of b_f; all other parameters are at baseline. (b) For endogenous-demand runs with $A_{f0} = 0.6\,\text{ha}$, $q_0 = 0.9$, $\tau_g = 20\,\text{yr}$, and $\tau_p = 25\,\text{yr}$, the table reports the initial population P_0 at which the minimum A_c comes within the stated distance of the floor $A_c^{\min} = 0.05\,\text{ha}$. The first two columns use a P_0 grid with step 0.1 ($\Delta t = 0.1\,\text{yr}$, $T = 400\,\text{yr}$); the final column uses bisection with the $10^{-6}\,\text{ha}$ proximity criterion ($\Delta t = 0.05\,\text{yr}$, $T = 1000\,\text{yr}$). In the tested runs, sampled drawdown depth is monotone across the P_0 grid (Fig. 1c); neither the finite grid nor selected bisections establish general continuity. These values define proximity bands, not a single threshold.

(a) Fast-land yield sweep (all rows composition-premium, $\Delta b_{\text{conv}} > 0$)

b_f (gha ha^{-1})	$\Delta b_{\text{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.804
0.85 (baseline)	+0.784	−0.0537	−0.998	1.138
1.20	+1.134	−0.0547	−1.014	1.606
2.00	+1.934	−0.0556	−1.029	2.675

(b) Initial population P_0 (cap) at which $\min_t A_c$ approaches $A_c^{\min}$ to within:

A_{c0} (ha)	within 10^{-2} ha	within 10^{-4} ha	within 10^{-6} ha
0.40	≈ 1.8	≈ 2.1	≈ 2.41
0.70	≈ 2.2	≈ 2.4	≈ 2.70
1.00	≈ 2.6	≈ 2.8	≈ 2.98
1.18	≈ 2.8	≈ 3.0	≈ 3.14

3.3. Technology gains are finite; debt depends on the shortfall

The technology boost and debt both change fast-land yield b_f, but in different ways. The boost has a fixed ceiling. Debt can accumulate while a shortfall persists, then approaches a finite level if that shortfall settles.

Proposition 3 (The technology boost is capped; debt depends on the shortfall)**.** *Under* (12)–(14)*:*

(i) The technology factor $(b_{f0}+T_b(t))/b_{f0}$ *stays between* 1 *and* $1+\Delta b/b_{f0}$*. Its log-scale gain is therefore bounded by* $\ln(1+\Delta b/b_{f0})$ *for all time (*0.78 *at baseline).*

(ii) Debt satisfies $D(t) \le \max\{D(0),\ \sup_{s\le t}[E-B]_+(s)/\eta\}$*. If the shortfall settles at* S^**, then* D *tends to* S^*/η*. The resulting reduction in fast-land yield depends on the size of that sustained shortfall.*

(iii) The debt-related yield loss exceeds the largest possible technology gain once the sustained shortfall exceeds $S^\dagger = (\eta/\alpha)\ln(1+\Delta b/b_{f0})$*, which is* 1.30 *gha at baseline.*

For the additive comparator $b_f = b_{f0} + T_b - \alpha_{\text{add}} D$*, where* α_{add} *is a distinct coefficient with units* $\text{ha}^{-1}\,\text{yr}^{-1}$*, the two effects enter symmetrically; the comparison in (iii) then reduces to* Δb *versus* $\alpha_{\text{add}} D^*$*. The asymmetry above follows from the model's multiplicative form.*

Proof. (i) is $0 \le T_b \le \Delta b$. (ii) From (13), $\dot D \le \bar S - \eta D$ with $\bar S = \sup_{s\le t}[E-B]_+$, so $D(t) \le \max\{D(0), \bar S/\eta\}$ by comparison; if $[E-B]_+ \to S^*$ then $D \to S^*/\eta$. (iii) compares $\alpha S^*/\eta$ with $\ln(1+\Delta b/b_{f0})$. □

With fixed demand and a persistent shortfall, D approaches S^*/η; larger shortfalls imply higher debt. At baseline, the asymptotic sustained-shortfall crossover is $S^\dagger \approx 1.3$ gha. It is a scenario-specific comparison, not a transient trajectory threshold: the relative effect along a run depends on $D(0)$, demand history, and the summary window. Debt nevertheless reduces fast-land yield over time.

A technology boost can raise B and carrying capacity; delayed population growth may raise demand too. Peak debt therefore depends on initial population and the summary window. In the matched SI §S8 experiment, for $P_0 = 1.0$ cap, peak debt over 0–250 yr is 0.02020 gha yr with the boost and 0.00813 gha yr without it (ratio 2.49). For $P_0 = 1.2$ cap, the full-run maximum is lower with the boost (0.05104 versus 0.06122 gha yr), but the maximum after the 40 yr wave midpoint is higher (0.02518 versus 0.01085 gha yr; ratio 2.32). These are conditional model results, not forecasts or empirical estimates.

3.4. Recovery requires an explicit, gated restoration flow

Conversion moves land from capital to fast use. The model has no route back to capital land unless restoration R_{rc} is active; fast-land retirement returns land to reserve instead. Lowering population, footprint, or shortfall therefore does not, by itself, reverse earlier conversion. With $\chi_r = 0$, A_c stays at its post-conversion level.

In the projected delayed-quality run ($A_{c0} = 1.0\,\mathrm{ha}$, $P_0 = 2.0\,\mathrm{cap}$, $\chi_r = 0.10\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $T = 400\,\mathrm{yr}$), A_c falls to about $0.398\,\mathrm{ha}$ and then recovers to $A_c(T) = 0.677\,\mathrm{ha}$. The recovery fraction is $\mathcal{R}_c = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}}) = 0.46$, matching the delayed-quality recovery table in the Supplementary Information (Table S5; $\Delta t = 0.1\,\mathrm{yr}$). Refining this same case to $\Delta t = 0.00078125\,\mathrm{yr}$ gives $\min A_c = 0.401217\,\mathrm{ha}$ and $A_c(T) = 0.678295\,\mathrm{ha}$; recovery remains partial (Section 3.6; SI §S10).

The two restoration gates operate under different conditions. Surplus-gated restoration is inactive during a shortfall, but may continue after a surplus opens while population adjusts. Reaching a specified endpoint therefore requires a target-limited rule (SI §S4.4). Deficit-gated restoration acts during shortfall, when supply is already insufficient. At the tested rates, conversion outpaces restoration and drawdown continues. The result depends on timing and rates; neither gate is universally preferable.

3.5. What independent observations identify: sharp static bounds

The ambiguity in Proposition 1(iii) can be quantified without introducing another dynamic experiment. Write $u = \Upsilon_c(q)$. Suppose total area $L = A_{tot}$, the minimum $a_{\min} = A_c^{\min}$, and the current fast-land yield b_f are known; assume independently justified, continuously attainable yield limits $0 < u_- \le u \le u_+ < b_f$. The only allocation restrictions used here are $A_f, A_r \ge 0$, $A_c \ge a_{\min}$, and $A_f + A_c + A_r = L$. The yield interval must be attainable independently of the permitted areas; additional joint restrictions require a different feasible set.

Proposition 4 (Sharp static identification intervals)**.** *For a feasible observed B, the exact set of compatible capital-land areas using B alone is*

$$\mathcal{I}_B = \left[a_{\min},\ \min\left\{ L, \frac{B}{u_-}, \frac{b_f L - B}{b_f - u_+} \right\} \right]. \tag{20}$$

If A_f is also independently known, put $C^{\mathrm{mod}} = B - b_f A_f$. For $C^{\mathrm{mod}} \geq 0$, the exact set is

$$\mathcal{I}_{B,A_f} = \left[\max\left\{ a_{\min}, \frac{C^{\mathrm{mod}}}{u_+} \right\}, \ \min\left\{ L - A_f, \frac{C^{\mathrm{mod}}}{u_-} \right\} \right]. \tag{21}$$

An interval with lower endpoint above its upper endpoint is infeasible; $C^{\mathrm{mod}} < 0$ is also infeasible. Independently knowing q reduces the latter set to $A_c = C^{\mathrm{mod}}/\Upsilon_c(q)$ if this value satisfies the allocation restrictions. These are sharp static sets, not sets of dynamically reachable states.

Proof. At fixed $A_c = a$, varying the admissible u and A_f gives exactly $u_- a \leq B \leq b_f(L - a) + u_+ a$. Solving these inequalities gives (20). When A_f is fixed, $u_- a \leq C^{\mathrm{mod}} \leq u_+ a$ gives (21). Continuity and attainment of the yield interval supply a feasible allocation at each permitted a; SI §S9 gives the construction and endpoint witnesses. □

For a model-scale illustration, take $b_f = 0.85\,\mathrm{gha\,ha^{-1}}$, $L = 2.5\,\mathrm{ha}$, $a_{\min} = 0.05\,\mathrm{ha}$, $u \in [0.05, 0.066]\,\mathrm{gha\,ha^{-1}}$, and $B = 0.825\,\mathrm{gha}$. The baseline yield law attains this entire interval on $q \in [0, 1]$. If an independent observation gives $A_f = 0.785/0.85 = 0.923529\ldots\,\mathrm{ha}$, then $C^{\mathrm{mod}} = 0.04\,\mathrm{gha}$. Table 3 shows how the additional observations narrow the compatible set. Even the middle interval crosses an arbitrarily chosen 0.7 ha reporting cut; that cut is not an ecological threshold. Independently observing $q = 0.9$ identifies $A_c = 0.717360\ldots\,\mathrm{ha}$ in this example. Observing only Υ_c need not identify q, because this yield law is not one-to-one. The illustration is not an estimate from NFA accounts. Uncertain yields or measurements require projection of a coherent joint uncertainty set, rather than treating these deterministic intervals as confidence intervals.

Table 3: Compatible capital-land area under nested observation packages in the illustrative static model. Bounds and widths are in ha; A_f and q in the last two rows are independent observations with the values specified in the text.

Observations	Lower	Upper	Width
B	0.050000	1.658163	1.608163
B, A_f	0.606061	0.800000	0.193939
B, A_f, q	0.717360	0.717360	0.000000

3.6. Integrated model contribution budgets

A positive conversion component does not, by itself, establish the net rise in B. Extending (16) to active restoration and varying fast-land yield gives, almost everywhere along the projected trajectory,

$$\dot{B} = \underbrace{(b_f - u)u_c}_{\text{conversion}} + \underbrace{b_f(\mu A_r - \eta_f A_f)}_{\text{maintenance/retirement}} + \underbrace{uR_{rc}}_{\text{restoration}} + \underbrace{b_{G,c}G_c'(q)A_c\dot{q}}_{\text{quality readout}} + \underbrace{A_f\dot{b}_f}_{\text{fast yield}} , \tag{22}$$

where $\dot{b}_f = e^{-\alpha D}\dot{T}_b - \alpha b_f \dot{D}$. Each displayed term has units gha yr^{-1}; its time integral has units gha. The last channel can therefore be split into technology and debt terms without changing the identity.

The five continuous-time contribution terms are integrated by trapezoidal quadrature on the numerical state nodes. Endpoint account change is evaluated separately, and the difference between the endpoint change and the sum of integrals measures numerical closure error. Seven nested timesteps from 0.05 to 0.00078125 yr assess this error (SI §S10).

Table 4 reports the fixed-demand simulation over 100 yr and the endogenous-demand restoration simulation over 400 yr. Under fixed demand, conversion contributes 0.547925 gha, offset principally by −0.311098 gha from maintenance and retirement; the endpoint increase is 0.231310 gha. In the restoration condition, the direct restoration contribution is 0.016561 gha, compared with 0.513933 gha from conversion and −0.218037 gha from maintenance and retirement. These signed contributions describe the accounting decomposition along a coupled trajectory. They do not estimate empirical causal effects. The quality contribution is negative because $G_c(q)$ decreases as q approaches the imposed upper bound from the specified initial value; the sign reflects the observation function rather than a decline in the quality state.

The finest-step residuals are 8.04×10^{-5} and 1.97×10^{-4} gha, respectively (0.0348% and 0.0628% of the endpoint changes). Closure improves under refinement but is not exact at finite step. SI §S10 reports the matched control and technology conditions, separates technology and debt contributions, and documents the state constraints. These calculations characterize the stated histories and parameters.

3.7. NFA account composition, 1961–2023

For the primary empirical analysis, we use complete world observations in NFA Edition 2026, results and metadata version 1.0, and exclude the

Table 4: Integrated contributions to model B, in gha, at $\Delta t = 0.00078125$ yr. The closure residual is $B(T) - B(0)$ minus the sum of independently integrated terms; it is a numerical discrepancy, not an additional mechanism. The fixed-demand condition has no restoration; the restoration condition has endogenous demand and surplus-gated restoration.

Contribution or statistic	Fixed demand	Restoration
Conversion	+0.547925	+0.513933
Maintenance/retirement	-0.311098	-0.218037
Restoration	+0.000000	+0.016561
Quality readout	-0.006470	-0.002577
Fast-yield change	+0.000873	+0.002687
Sum of integrals	+0.231230	+0.312567
$B(T) - B(0)$	+0.231310	+0.312764
Closure residual	+0.000080	+0.000197

estimated 2024–2025 values. Our 1961–2022 sensitivity uses the same account edition and population series.

Between 1961 and 2023, total biocapacity account value B rises from 9.7254×10^9 to 12.1428×10^9 gha (24.86%; $\Delta \ln B = +0.221997$). Population grows from 3.06487 to 8.09173 billion. Over the same interval, B/P falls from 3.17319 to 1.50065 gha person^{-1} (52.71%; $\Delta \ln(B/P) = -0.748840$). The Global Ecological Footprint of Production per person (E_{prod}/P) rises from 2.19700 to 2.56462 gha person^{-1} (16.73%). Meanwhile, E_{prod}/B rises from 0.69236 to 1.70901, first exceeding one in 1973.

The exact identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ yields $+0.970837 - 0.748840 = +0.221997$. It separates population-scale change from change in biocapacity per person, but does not explain their causes. The same-edition 1961–2022 sensitivity gives $\Delta \ln B = +0.219497$, a 52.41% fall in B/P, a 17.70% rise in E_{prod}/P, and 1973 as the first year with $E_{\text{prod}}/B > 1$ (SI §S5).

The component series show how NFA account composition changed (Fig. 3; Table 5). Cropland contributes +24.06 percentage points to the aggregate's +24.86% change—about 96.8% of the net percentage change. Forest biocapacity contributes −5.95 points, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland account sum, which includes built-up land, is nearly unchanged (1.0092×, or +0.80 points) because the built-up increase offsets declines elsewhere. These are changes in account values. They do not establish changes in forest biomass, soil

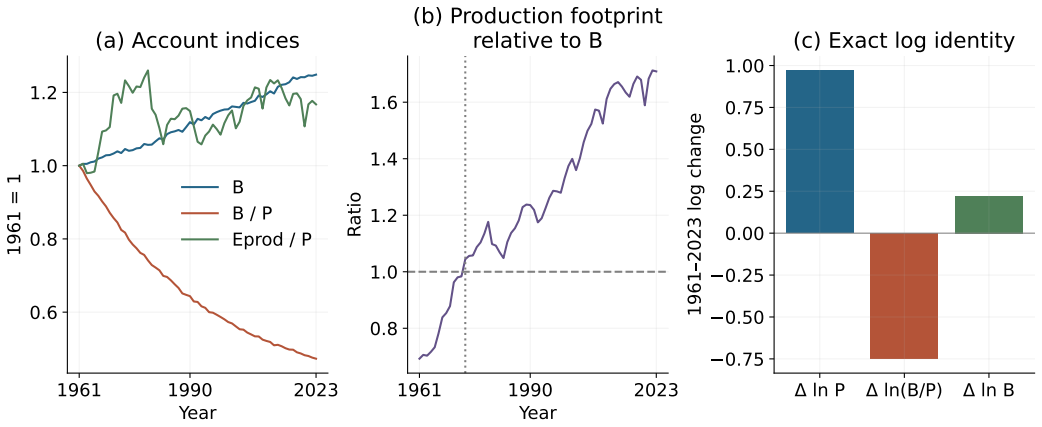


Figure 2: Aggregate and per-capita movements in NFA Edition 2026 (world, 1961–2023; complete observations). B is total biocapacity, E_{prod} is the workbook's Global Ecological Footprint of Production, and P is the single UN PopStat population series; per-capita values are calculated from totals. (a) Indices (1961 = 1) of B, B/P, and E_{prod}/P. (b) E_{prod}/B, first above one in 1973. (c) Exact log identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$. Account values are in gha; plotted indices, ratios, and log changes are dimensionless. The 2024–2025 estimated rows are excluded.

condition, wetland extent, fish biomass, or model area A_c. Built-up land is an account class, not ecological capital. Component values sum to total biocapacity within workbook rounding at both endpoints.

We also use external cropland area and cereal yield to build an illustrative proxy for the fast-land contribution (SI §S5). Anchoring it to the measured 1961 NFA cropland share ($\phi_0 = 0.128647$, about 12.86%), the exact log-change split assigns +0.268662 to the cereal-yield index, +0.037710 to cropland area, and −0.084375 to the residual. Together these sum to $\Delta \ln B = +0.221997$. The identity is exact for this proxy, but its allocation changes with the anchor and is not a physical-stock attribution.

3.8. Falsifiable predictions

The model yields five conditional predictions, with an operational test map in SI §S11. Aggregate B alone cannot test them: the composition mechanism requires independent observations or defensible proxies for A_c, fast-land area, and yield; testing delay, recovery, and demand-threshold predictions also requires relevant time series and intervention or demographic data.

Table 5: NFA Edition 2026 biocapacity-account endpoint ratios and contributions. Contributions are $100\,[B_i(T) - B_i(1961)]/B(1961)$ percentage points; endpoint ratios are $B_i(T)/B_i(1961)$. The 2022 sensitivity is from the same edition. All entries are account values, not physical-stock measurements. Individually rounded contributions need not sum to the independently rounded total.

Component	2023 ratio	2023 contribution (pp of B_{1961})	2022 sensitivity ratio	2022 contribution (pp of B_{1961})
Cropland	2.8701	+24.0577	2.8514	+23.8183
Grazing land	0.9828	−0.2688	0.9815	−0.2890
Forest biocapacity	0.8996	−5.9525	0.9019	−5.8137
Fishing grounds	0.9627	−0.4212	0.9596	−0.4559
Built-up land†	8.7935	+7.4416	8.6298	+7.2854
Non-cropland sum†	1.0092	+0.7990	1.0083	+0.7267
Total biocapacity	1.2486	+24.8568	1.2454	+24.5450

†Built-up land is not ecological capital; the non-cropland account sum includes it. Component sums match total biocapacity within workbook rounding.

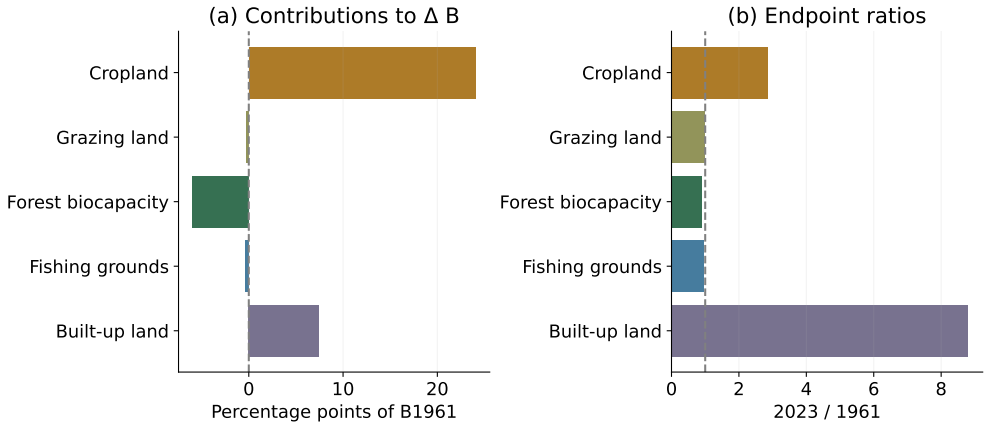


Figure 3: NFA Edition 2026 land-type biocapacity-account changes, world, 1961–2023. (a) Component contributions in percentage points of B_{1961}. (b) Endpoint ratios $B_i(2023)/B_i(1961)$. "Forest biocapacity" is the workbook's exact label. All bars are account values, not measurements of physical stocks; built-up land is an NFA accounting class, not ecological capital.

P1. **Conversion can raise the total under a specific condition.** In the baseline fixed-demand run, $\dot{A}_c < 0$ while $\dot{B} > 0$ for a finite interval (Fig. 1a). Conversion adds to B exactly when $\Delta b_{\text{conv}} > 0$, although other flows, quality, debt, technology, and restoration determine the net change. With baseline shares $\sigma_f = \sigma_c = 1$, conversion requires demand to exceed total biocapacity ($R_B = E/B > 1$). The reserved-share extension can permit conversion when $R_B \leq 1$, but that is not a baseline result. A composition mask is rising B with falling A_c; a liquidation mask is steady E with falling A_c. The two patterns may overlap but are not interchangeable. The sign of Δb_{conv} describes conversion's contribution, not necessarily the net change in B.

P2. **The duration depends on model rates, not a universal horizon.** The slow return rate λ_1 (along with τ_{conv}, μ, η_f, and b_c/b_f) sets the timescale; Δb_{conv} and the initial shortfall affect the size of the change. In the baseline, most conversion occurs within one to two e-folding times, when the remaining gap shrinks by a factor of e.

P3. **The reduced model has no fold-type early warning.** Return time does not grow as demand approaches the ceiling, and drawdown depth changes continuously with demand (Fig. 1c). The model instead signals that A_c is approaching its minimum. It predicts no fold-related warning before that boundary.

P4. **The baseline 20/40-year comparison is saturation-induced.** At $q(s) = 0.9$, $\rho_c = 0.08\,\text{yr}^{-1}$ and $q_{\max} = 1$, the projected quality reaches its imposed ceiling at $125/9 \approx 13.8889\,\text{yr}$, before either $\tau_g = 20$ or $40\,\text{yr}$ elapses. Those quality paths and matched outcomes therefore coincide; the $\tau_g = 5\,\text{yr}$ case is distinct. The finite comparison does not establish general delay robustness (SI Table S2).

P5. **Recovery requires a return flow.** Lower demand alone does not restore capital-land area. The model needs an explicit restoration flow. A surplus-gated flow is inactive during shortfall, whereas a deficit-gated flow can further reduce provision.

Empirical test design.. P1 requires an independently mapped area transition and yield contrast; rising account totals alone do not establish its conversion mechanism. P2–P3 require local response experiments or identified interventions across demand levels, not a trend test on global B. P4 tests the

specified projection and history against an exact path; its failure in ecological data would reject that mapping, not establish a universal delay threshold. P5 requires measured capital-area return flows and matched demand changes. The SI specifies independent observables, fitted quantities, conditional nulls, rejection outcomes, and uncertainty treatment before validation data are inspected. None of the NFA accounting identities is presented as an independent test of these dynamics.

4. Discussion

4.1. What the two-land model shows

The two-land model yields three results relevant to human–environment models (Schaefer, 1954; May, 1973; Meadows et al., 1972, 2004; Wackernagel and Rees, 1996; Brander and Taylor, 1998). First, conversion can add to total biocapacity while A_c falls. Whether B rises overall also depends on other land flows and quality change; Proposition 1 gives the exact condition for the conversion term. The technology boost is capped, while debt depends on the magnitude and duration of shortfall (Proposition 3).

Second, under the reduced fixed-demand assumptions ($q = q_{\max}$, $D = 0$, technology boost off, and no restoration), the equality balance relation declines steadily with A_c until it meets the area floor. It does not fold back. The absence of a fold or critical slowing applies to this equality relation, not to every trajectory or to the separate surplus no-conversion family. With one-way conversion and no return flow, A_c remains below its pre-overshoot level after conversion stops, although it no longer declines at the equality point (Table 6). Third, recovery requires an explicit return flow; its effect depends on whether the restoration gate opens during a shortfall or a surplus.

SI §S7 gives a separate one-stock comparator. Its state A has comparator-specific units; it is not A_c. With constant parameters and no debt, its equilibrium relation is $B(A) = (b + b_G\rho)A - (b_G\rho/A_{\max})A^2$, with $P = B(A)/e$. At the illustrated settings, $b_G\rho < b$, so the relation is monotone; an interior maximum would require other parameters. Its neutral equilibrium family and selected delay trajectories belong only to that formulation. They establish neither a fold nor a universal critical delay or collapse mechanism in the two-land model.

These results also clarify when to expect common early-warning indicators. Rising autocorrelation and variance are associated with approaching a fold (Scheffer et al., 2001). This reduced model reaches a minimum-area boundary instead, so those indicators need not rise first. Their absence would not establish safety. The relevant model signal is the path of A_c, which the aggregate alone cannot reveal.

Table 6: Equilibrium, trajectory, and tested delay-sensitivity properties under the stated assumptions.

Property	Finding	Scope
Fold on the reduced fixed-demand branch	Absent	$B_{eq}(A_c)$ is strictly monotone; the balance point collides with the typed floor at $E_{\text{ceil}} \approx 1.138\,\text{gha}$ (Proposition 2)
Critical slowing as $E \to E_{\text{ceil}}$	Absent	The slow eigenvalue remains negative and constant along the reduced branch
Hysteresis of the reduced equality relation	Absent on this relation	There is at most one feasible equality point; fixed-demand surplus no-conversion allocations form a separate family, and the deficit-side eigenvalues are one-sided
Path dependence of realized A_c	Present, conditional	With $R_{rc} = 0$, one-way conversion leaves A_c below its pre-overshoot level after shortfall closure, while $\dot{A}_c = 0$ at equilibrium
Discontinuous response to endogenous demand	Not observed	The sampled minimum A_c varies monotonically across tested initial populations; proximity thresholds depend on tolerance. The finite runs do not establish a general continuity theorem
Regeneration-delay comparison (τ_g, with $\tau_p = 25\,\text{yr}$)	Degenerate at 20/40 yr	Baseline quality reaches its imposed ceiling before either delay elapses, so those paths coincide; $\tau_g = 5\,\text{yr}$ is distinct. This is not general delay robustness (SI Table S2)

4.2. Measurement implications

Biocapacity in gha is a value-weighted total, not an inventory of land classes. A 20% increase cannot distinguish modest gains across all components from a large gain in one that offsets losses elsewhere. Proposition 1(iii) formalizes the limit: even with fixed per-hectare yields, the total and the conserved-area constraint leave multiple feasible allocations among fast, capital, and reserve land.

A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Sections 2 and 4 of Abaee, 2026b). The model total B is also a linear sum, but the companion analyzes a different, finite-plan setting. There, the compensatory protocol accepts states even when no single available plan meets every component floor (Theorem 2 in Abaee (2026b)). Theorem 4(iii) in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe; thus no positive-elasticity member is uniformly safe on that region. If feasibility is required throughout a transition, falling below a land-class minimum remains a violation even if the endpoint later recovers.

For the NFA, these findings concern account composition, not physical stocks. In Edition 2026, total world biocapacity account value B rises 24.86% from 1961 to 2023, while B/P falls 52.71%. The Global Ecological Footprint of Production per person (E_{prod}/P) rises 16.73%, and E_{prod}/B first exceeds one in 1973. B and E_{prod} are annual account amounts in gha, with the annual basis implicit; per-person values are in gha per person, and E_{prod}/B is dimensionless.

Cropland contributes +24.06 percentage points to the total change; Forest biocapacity contributes −5.95, grazing land −0.27, fishing grounds −0.42, and built-up land +7.44. The non-cropland sum is nearly flat because these account movements offset one another, including the large built-up-land increase. SI §S5 reports the same-edition 1961–2022 comparison. None of these values measures forest biomass or condition, soil carbon, wetland extent, fish biomass, or another physical stock. In particular, a decline in Forest biocapacity does not establish a decline in forest biomass. Built-up land is an NFA accounting class, not ecological capital. The ratios, contributions, vintage, and endpoint sensitivity describe account composition only (Lo et al., 2026; Borucke et al., 2013; Blomqvist et al., 2013; Giampietro and Saltelli, 2014; Galli et al., 2016; van den Bergh and Grazi, 2015).

The model result is distinct from the NFA trends. In the fixed-demand baseline (Fig. 1a), A_c falls from 1.500 to 0.807659 ha (46.16%), while model B rises from 0.594 to 0.825 gha (39%). Quality q rises from 0.9 to 1.0, moving farther above $q^* = q_{\max}/2$, where $\Upsilon_c(q)$ peaks. Its per-hectare value therefore falls from 0.05576 to 0.05000 gha ha^{-1} (10.33%), and the modeled contribution $\Upsilon_c(q)A_c$ falls from 0.08364 to 0.04038 gha (51.72%).

With restoration off, one-way conversion leaves A_c below its starting value after the shortfall closes. It then remains at the lower balance level rather than declining indefinitely. This is a conditional model result, not an observation or estimate of global physical-stock change.

We assembled no common-boundary global physical-stock series for these components and therefore report no physical-stock composite or global decline finding. The SI measurement specification is a specification—not an inventory—with candidate stock, extent, condition, and annual-flow measures and the requirements for comparing them; it contains no endpoint values. Until spatial and temporal boundaries, units, uncertainty, overlap, valuation or normalization, and permitted substitution are specified, these unlike measures should remain a nonadditive vector. SEEA Ecosystem Accounting also reports ecosystem extent and condition separately from ecosystem services and monetary presentations (United Nations et al., 2021).

The external cropland-area and cereal-yield series provide an illustrative proxy, not direct measurements of the model's land classes. The proxy is anchored to the 1961 NFA cropland share; cereal yield is a productivity indicator, not an all-crops measure, and the area and account boundaries do not match by construction. External cropland area A_f is in hectares and index-scaled yield $\hat{b}_f$ in gha ha^{-1}, so $X = \hat{b}_f A_f$ and $C^{\text{emp}} = B - X$ are account amounts in gha. The residual can absorb boundary and calibration differences, other account components, and conversion-factor changes. It is neither $\Upsilon_c(q)A_c$ nor a physical-stock measure. Physical interpretation would require independent, boundary-consistent measures of area, condition, yield, and regeneration, together with a validated mapping and valuation. Forest area could inform one part of the effective capital class, but cannot represent the entire class or its condition.

4.3. Policy controls and limits

The conversion-rate law in Eq. (5) preserves the capital-land floor from safe initial states. For $y = A_c - A_c^{\min} \geq 0$, $0 \leq u_c \leq y/\tau_{\text{conv}}$ and $R_{rc} \geq 0$ imply $\dot{y} \geq -y/\tau_{\text{conv}}$, hence $y(t) \geq y(0)e^{-t/\tau_{\text{conv}}}$. The floor is approached

asymptotically; a stop is an operational safeguard rather than an additional continuous-time guarantee. A ceiling on fast-land area constrains A_f, whereas the capital-land minimum concerns A_c; the two restrictions are distinct. Constraint preservation also differs from state identification. The constant choice $u_c = 0$ preserves the capital-land floor without observing A_c, but does not generally meet a provision objective. Policies that depend on remaining capital-land area require observations or valid bounds for A_c beyond aggregate B. Joint provision and area objectives therefore need to be assessed under specified admissible controls and state information.

The restoration gates also behave differently. Surplus-gated restoration is inactive during a shortfall; deficit-gated restoration acts during the shortfall and can reduce provision further. This comparison echoes the distinction between protective and mobilizing institutional feedback in related work (Abaee, 2026a). With a common reserved share, $\sigma_f = \sigma_c = \sigma < 1$, conversion is inactive while $E \leq \sigma B$. With separate shares, the condition is $E \leq \sigma_f Y_f + \sigma_c Y_c$. This reservation applies to annual human-available flow, not to spatially protected area. Half-Earth, by contrast, is a spatial conservation target (Wilson, 2016). The connection is only the abstract idea of reservation: the model neither maps protected areas nor evaluates that proposal. Robustness to disturbances and parameter uncertainty remains untested.

4.4. Relation to prior work

The ledger companion distinguishes weak and strong sustainability as operating regimes according to whether material loops close at throughput. In its terms, a positive weighted sum ranks outcomes under weak sustainability, whereas componentwise adequacy is the strong-sustainability predicate (Abaee, 2026c, Sections 1.1 and 10.1). The model total B is an annual biocapacity-account total. The link is an analogy about compensation, not an identity with the companion's material-flow ledger or a complete sustainability classification. The assessment companion studies aggregate measures and typed transition criteria in a separate finite-plan setting (Abaee, 2026b); the governance companion studies institutional feedback (Abaee, 2026a). The delays here represent ecological regeneration and demographic response, not institutional review.

4.5. Limitations

Several limitations set the scope of the results. Ecological-capital land combines unlike assets, so its single quality state, projected regeneration rule, composition diagnostic, and demand ceiling are effective model quantities—not estimates for forests, soils, or wetlands. The quality index is uncalibrated; $q_{\max} = 1$ is an imposed admissibility bound, and ρ_c and τ_g are scenario parameters rather than estimates from recovery studies. Most other parameters are representative rather than fitted. A cross-section cannot identify b_c, $b_{G,c}$, and ρ_c separately. Recovery records from different systems and external measurement frameworks are precedents for future ecosystem-specific measurement and modeling only; they do not validate the present equation or parameter values. The records describe regeneration responses, not the delay τ_g. We hold the per-person footprint constant to isolate feedbacks among land, biocapacity, and demand. A changing footprint would require another model extension.

The composition pattern depends on demand and on the mechanisms included here; under the stated assumptions, it is bounded. The exact shortfall switch makes stability at $S = 0$ one-sided. Drawdown thresholds are reported as proximity bands because they depend on tolerance and time step. We fix debt repayment at $\eta = 0.05\,\mathrm{yr}^{-1}$; the zero-repayment limit ($\eta \to 0$), which removes the debt equilibrium, is not analyzed. NFA accounts combine area and yield and therefore cannot identify them separately. Finally, the current-quality comparator $G_c(q(t))$ is retained only as a sensitivity check; it is not the selected projected delayed-quality model. In the displayed finite comparison it lengthens the demonstration window by 1.3 yr and shifts the initial-population threshold by at most 0.001 cap (SI §S3).

5. Conclusions

Aggregate biocapacity alone generally does not identify changes in its underlying land allocation. In the two-land model, conversion raises total B at the margin when fast-land contribution per hectare exceeds capital land's effective value ($b_f > \Upsilon_c(q)$), while A_c falls if restoration is off (Proposition 1). Other model terms determine whether total B rises overall. This is a conditional accounting mechanism, not an empirical census of global physical stocks.

The reduced fixed-demand equality relation reaches the specified minimum-area boundary without a fold or critical slowing. This is a

statement about that relation, not the separate fixed-demand surplus no-conversion family or every trajectory of the delayed model. Realized paths remain history-dependent: when shortfalls subside and restoration is off, A_c remains below its pre-overshoot level, but it does not continue to decline at the balance point. In the representative baseline run, total model biocapacity rises 39% while ecological-capital area falls 46.16%, its per-hectare quality term $\Upsilon_c(q)$ falls 10.33%, and its modeled contribution $\Upsilon_c(q)A_c$ falls 51.72%.

Independent yield and land observations turn the static non-identification claim into explicit compatible-area intervals; the example narrows from $[0.05, 1.658163]$ to $[0.606061, 0.8]$ ha when fast-land area is independently known. The integrated budgets quantify conversion alongside the offsetting channels with measured finite-step closure errors. Neither result identifies physical capital from NFA account totals or establishes robustness to other delay histories.

Separately, NFA Edition 2026 world account values rise 24.86% between 1961 and 2023, while biocapacity per person B/P falls 52.71% and the production footprint per person rises 16.73%. Cropland contributes +24.06 percentage points, Forest biocapacity −5.95, and Built-up land +7.44 as an accounting class. These normalized annual account values are not observations of physical stocks or model A_c.

No common-boundary global physical-stock series was assembled, so we report no composite or global physical-stock decline finding. The SI measurement specification sets out candidate measures and comparison requirements; it is not an empirical inventory. Physical measures with different units and boundaries cannot be combined into one stock total without explicit treatment of uncertainty, overlap, valuation or normalization, and allowed substitution. The auxiliary proxy residual $C^{\text{emp}} = B - \hat{b}_f A_f$ is neither a physical-capital measure nor the model contribution $\Upsilon_c(q)A_c$; a future comparison would require independent observations and a validated mapping.

Component accounts and independent physical observations provide complementary information and should be reported alongside the aggregate. Returning to the orchard metaphor, a ledger of annual harvest alone cannot tell whether the trees, roots, and soil that support future harvests have changed. That physical record requires independent, comparable observations; it cannot be inferred from the account total. A harvest total is not a census of the orchard.

Acknowledgments

The author thanks Global Footprint Network for making its data publicly available; Mathis Wackernagel for his thoughtful review; the editors of *Ecological Modelling* for their trust and support; and Arena.ai for its ingenuity and online tools.

Declarations

Data availability. Empirical input sources are cited in the manuscript. Model implementation, identification calculations, contribution integrals, timestep comparisons and figure files are available at https://github.com/MIKEAA2020/general-sustainability/tree/main/batch%209/agent%202/ecomod_audit_remediation_C18_A01. The package includes frozen NFA Edition 2026 and OWID inputs with attribution and hashes, a 1961–2023 annual joined series, executable endpoint/component/proxy transformations, and figure-generating code. The supplementary drawdown, gate-sign and one-stock experiments have versioned code and numerical outputs. Source versions and file hashes are provided in the accompanying metadata; the NFA workbook and NFA-derived data retain CC BY-SA 4.0 attribution.

Funding. No funding was received.

Competing interests. The author declares no competing interests.

CRediT authorship contribution statement. **Amin Abaee:** Conceptualization; Methodology; Software; Formal analysis; Writing – original draft; Writing – review & editing.

Declaration of generative AI in scientific writing. DeepSeek AI and Claude (Anthropic) assisted with computational-reproducibility checks under the author's supervision; the author reviewed and edited the content and takes full responsibility for the publication.

References

Abaee, A. (2026a). *Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter*. Working paper. Manuscript.

Abaee, A. (2026b). *Aggregation is a claim, not a presentation: the geometry of the acceptance gap between compensatory and noncompensatory transition assessment*. Working paper. Manuscript.

Abaee, A. (2026c). *Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons*. Working paper. Manuscript.

Anderies, J. M. (2000). On modeling human behavior and institutions in simple ecological economic systems. *Ecological Economics*, 35(3), 393–412.

Becker, W., Saisana, M., Paruolo, P. & Vandecasteele, I. (2017). Weights and importance in composite indicators: Closing the gap. *Ecological Indicators*, 80, 12–22.

Blomqvist, L., Brook, B. W., Ellis, E. C., Kareiva, P. M., Nordhaus, T. & Shellenberger, M. (2013). Does the shoe fit? Real versus imagined ecological footprints. *PLoS Biology*, 11(11), e1001700.

Borucke, M., Moore, D., Cranston, G., Gracey, K., Iha, K., Larson, J., Lazarus, E., Morales, J. C., Wackernagel, M. & Galli, A. (2013). Accounting for demand and supply of the biosphere's regenerative capacity: The National Footprint Accounts' underlying methodology and framework. *Ecological Indicators*, 24, 518–533.

Brander, J. A. & Taylor, M. S. (1998). The simple economics of Easter Island: A Ricardo–Malthus model of renewable resource use. *American Economic Review*, 88(1), 119–138.

Calvin, K. & Bond-Lamberty, B. (2018). Integrated human–Earth system modeling — state of the science and future directions. *Environmental Research Letters*, 13(6), 063006.

Cohen, J. E. (1995). Population growth and Earth's human carrying capacity. *Science*, 269(5222), 341–346.

Dasgupta, P. (2021). *The Economics of Biodiversity: The Dasgupta Review*. London: HM Treasury.

Donges, J. F., Winkelmann, R., Cornell, S. E., Lucht, W., Dyke, J. G., Rockström, J., Heitzig, J. & Schellnhuber, H. J. (2017). Closing the loop:

Reconnecting human dynamics to Earth system science. *The Anthropocene Review*, 4(2), 151–157.

Erb, K.-H., Luyssaert, S., Meyfroidt, P., Pongratz, J., Don, A., Kloster, S., Kuemmerle, T., Fetzel, T., Fuchs, R., Hasegawa, T., Haberl, H., et al. (2017). Land management: data availability and process understanding for global change studies. *Global Change Biology*, 23(2), 512–533.

Fischer, S. M., Joy, M. K., Abrahamse, W., Milfont, T. L. & Petherick, L. M. (2022). The use and misuse of composite environmental indices. *bioRxiv*, 2022.03.15.484501. https://doi.org/10.1101/2022.03.15.484501.

Global Footprint Network (2009). *Ecological Footprint Standards 2009*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/Ecological_Footprint_Standards_2009.pdf.

Global Footprint Network (2019). *Working Guidebook to the National Footprint Accounts 2019*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/National_Footprint_Accounts_Guidebook_2019.pdf.

Galli, A., Giampietro, M., Goldfinger, S., Lazarus, E., Lin, D., Saltelli, A., Wackernagel, M. & Müller, F. (2016). Questioning the ecological footprint. *Ecological Indicators*, 69, 224–232.

Giampietro, M. & Saltelli, A. (2014). Footprints to nowhere. *Ecological Indicators*, 46, 610–621.

Haberl, H., Erb, K.-H., Krausmann, F., Gaube, V., Bondeau, A., Plutzar, C., Gingrich, S., Lucht, W. & Fischer-Kowalski, M. (2007). Quantifying and mapping the human appropriation of net primary production in earth's terrestrial ecosystems. *PNAS*, 104(31), 12942–12947.

Henderson, K. & Loreau, M. (2018). How ecological feedbacks between human population and land cover influence sustainability. *PLoS Computational Biology*, 14(8), e1006389.

Hutchings, J. A. & Reynolds, J. D. (2004). Marine fish population collapses: consequences for recovery and extinction risk. *BioScience*, 54(4), 297–309.

Krausmann, F., Erb, K.-H., Gingrich, S., Haberl, H., Bondeau, A., Gaube, V., Lauk, C., Plutzar, C. & Searchinger, T. D. (2013). Global human appropriation of net primary production doubled in the 20th century. *PNAS*, 110(25), 10324–10329.

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M. & Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0.* Prepared for Footprint Data Foundation by researchers at York University and the University of Iceland. Workbook: https://footprint.info.yorku.ca/files/2026/04/NEFBA_Data_global_national_ecological_footprint_biocapacity.xlsx; release notes: https://footprint.info.yorku.ca/files/2026/04/NEFBA_ReleaseNotes.pdf; data page: https://footprint.info.yorku.ca/data/.

May, R. M. (1973). *Stability and Complexity in Model Ecosystems.* Princeton: Princeton University Press.

Meadows, D. H., Meadows, D. L., Randers, J. & Behrens, W. W. (1972). *The Limits to Growth.* New York: Universe Books.

Meadows, D. L., Behrens, W. W., Meadows, D. H., Naill, R. F., Randers, J. & Zahn, E. (1974). *Dynamics of Growth in a Finite World.* Cambridge, MA: Wright-Allen Press.

Meadows, D. H., Randers, J. & Meadows, D. L. (2004). *The Limits to Growth: The 30-Year Update.* White River Junction, VT: Chelsea Green.

Meyer, P. S. (1994). Bi-logistic growth. *Technological Forecasting and Social Change*, 47(1), 89–102.

Meyer, P. S. & Ausubel, J. H. (1999). Carrying capacity: A model with logistically varying limits. *Technological Forecasting and Social Change*, 61(3), 209–214.

Motesharrei, S., Rivas, J., Kalnay, E., Asrar, G. R., Busalacchi, A. J., Cahalan, R. F., Cane, M. A., Colwell, R. R., Feng, K., Franklin, R. S., et al. (2016). Modeling sustainability: Population, inequality, consumption, and bidirectional coupling of the Earth and human systems. *National Science Review*, 3(4), 470–494.

Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A. & Giovannini, E. (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide.* Paris: OECD Publishing.

Neubauer, P., Jensen, O. P., Hutchings, J. A. & Baum, J. K. (2013). Resilience and recovery of overexploited marine populations. *Science*, 340(6130), 347–349.

Our World in Data (2026a). *Cereal yield.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cereal-yield.

Our World in Data (2026b). *Cropland area.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cropland-area.

Poeplau, C., et al. (2011). Temporal dynamics of soil organic carbon after land-use change in the temperate zone–carbon response functions as a model approach. *Global Change Biology*, 17(7), 2415–2427.

Poorter, L., et al. (2016). Biomass resilience of Neotropical secondary forests. *Nature*, 530(7590), 211–214.

Schaefer, M. B. (1954). Some aspects of the dynamics of populations important to the management of the commercial marine fisheries. *Bulletin of the Inter-American Tropical Tuna Commission*, 1(2), 25–56.

Scheffer, M., Carpenter, S., Foley, J. A., Folke, C. & Walker, B. (2001). Catastrophic shifts in ecosystems. *Nature*, 413(6856), 591–596.

Simon, J. L. (1996). *The Ultimate Resource 2.* Princeton: Princeton University Press.

United Nations et al. (2021). *System of Environmental-Economic Accounting—Ecosystem Accounting (SEEA EA).* White cover publication, pre-edited text subject to official editing. New York: United Nations. https://seea.un.org/ecosystem-accounting.

United Nations (2022). *World Population Prospects 2022: Summary of Results.* New York: Department of Economic and Social Affairs, Population Division.

van den Bergh, J. C. J. M. & Grazi, F. (2015). Reply to the first systematic response by the Global Footprint Network to criticism: a real debate finally? *Ecological Indicators*, 58, 458–463.

van den Bergh, J. C. J. M. & Rietveld, P. (2004). Reconsidering the limits to world population: Meta-analysis and meta-prediction. *BioScience*, 54(3), 195–204.

Wackernagel, M. & Rees, W. (1996). *Our Ecological Footprint: Reducing Human Impact on the Earth.* New Society Publishers.

Wilson, E. O. (2016). *Half-Earth: Our Planet's Fight for Life.* New York: Liveright.